

cannot be A.

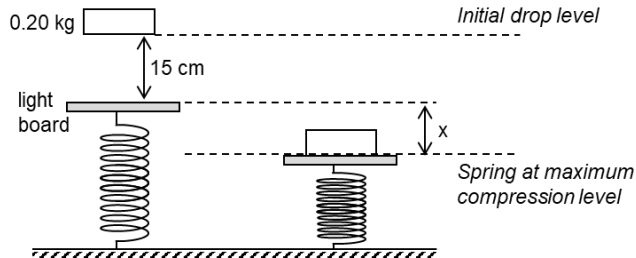
- 9** **D** No kinetic energy at the initial drop level as well as at the final maximum compression of spring.
Comparing the total energy at the initial and final positions,
gain in elastic potential energy = total loss in gravitational potential energy

$$\frac{1}{2}kx^2 = mg(0.15 + x)$$

$$\frac{1}{2}(85)x^2 = (0.20 \times 9.81)(0.15 + x)$$

$$42.5x^2 - 1.962x - 0.2943 = 0$$

$$x = 0.109 \text{ m}$$



- 10** **C** At maximum speed engine force = drag force of ky